

Lemma.

Let K/F be a field extension, and let $\alpha \in K$ be algebraic over F . Then, for any $\sigma \in \text{Aut}(K/F)$, $\sigma\alpha$ is a root of the minimal polynomial for α over F .

Proof. Let $f(x) = x^n + a_{n-1}x^{n-1} + \dots + a_1x + a_0 \in F[x]$ be the minimal polynomial for α . Then, $f(\alpha) = 0$, so $\sigma(f(\alpha)) = 0$. (The field homomorphism preserves zero).

Now, $0 = \sigma(f(\alpha)) = \sigma(\alpha^n + a_{n-1}\alpha^{n-1} + \dots + a_1\alpha + a_0)$

$$\begin{aligned} \sigma \text{ fixes } F \quad \hookrightarrow &= (\sigma\alpha)^n + \sigma(a_{n-1}) \cdot (\sigma\alpha)^{n-1} + \dots + \sigma(a_1) \cdot (\sigma\alpha) + \sigma(a_0) \\ &= (\sigma\alpha)^n + a_{n-1}(\sigma\alpha)^{n-1} + \dots + a_1(\sigma\alpha) + a_0 \\ &= f(\sigma\alpha). \end{aligned}$$

Prop. Let $H \leq \text{Aut}(K)$ be a subgroup of the group of all automorphisms of K .

Then, the collection F of elements of K fixed by all the elements of H is a subfield.

Proof. First, given $\sigma \in \text{Aut}(K)$, the set $F_\sigma = \{a \in K \mid \sigma(a) = a\}$ is subfield, because:

Let $a, b \in F_\sigma$ be given. Then, $\sigma(a-b) = \sigma(a) - \sigma(b) = a-b$, and if $b \neq 0$, then $\sigma(a \cdot b^{-1}) = \sigma(a) \cdot \sigma(b)^{-1} = a \cdot b^{-1}$, so the claim is proved.

And now, since the arbitrary intersection of fields is field, so proved. ~~QED~~
Such a subfield is called the fixed field of $S \subseteq \text{Aut}(K)$.

Prop. Let K be a field. Then,

If $F_1 \leq F_2 \leq K$, then $\text{Aut}(K/F_2) \leq \text{Aut}(K/F_1)$.

If $H_1 \leq H_2 \leq \text{Aut}(K)$, then $F_{H_2} \leq F_{H_1}$.

In other say, The inclusion relation of automorphisms and fixed field is reverse.

Prop. Let E be the splitting field over F of the polynomial $f(x) \in F[x]$.
 Then, $|\text{Aut}(E/F)| \leq [E:F]$, and the equality holds if $f(x)$ is separable over F .
 Proof. Using induction for $[E:F] = n$. If $n=1$, then $E=F$, so $\text{Aut}(E/F)$ is trivial.
 If $n>1$, then $f(x)$ has at least one irreducible factor $p(x)$ with $\deg p > 1$.
 Let $\alpha \in E$ be a fixed root of $p(x)$. By the Conjugate isomorphism Theorem,
 there are at most $\deg p$ extensions of identity $\varphi: F \rightarrow F$ to $\varphi: F(\alpha) \rightarrow E$,
 and the number of extensions is exactly $\deg p$ if $p(x)$ is separable.
 By the tower Lemma, $[E:F(\alpha)] \cdot [F(\alpha):F] = [E:F]$ and since $[F(\alpha):F] > 1$, so
 $[E:F(\alpha)] < n$, we can use the assumption of induction, (clearly, E is splitting field of f over $F(\alpha)$)
 we can deduce that $|\text{Aut}(E/F(\alpha))| \leq [E:F(\alpha)]$. Finally,

$$|\text{Aut}(E/F)| = \underbrace{|\text{Aut}(E/F(\alpha))|}_{\substack{\uparrow \\ \text{the possibilities of composition}}} \cdot \underbrace{|\text{Aut}(F(\alpha)/F)|}_{\substack{\uparrow \\ [F(\alpha):F] = \deg p}} \leq [E:F(\alpha)] \cdot [F(\alpha):F] = [E:F]$$

Def. A finite extension K/F is called a Galois Extension if:

$$|\text{Aut}(K/F)| = [K:F].$$

In such a case, denote $\text{Gal}(K/F) = \text{Aut}(K/F)$.

